

Pragadeesh KMS

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📍 Davis, CA

🌐 LinkedIn

🌐 Portfolio

EDUCATION

University of California, Davis

Jan 2026 – Present

M.S. Computer Science

SRM University, Chennai

Sept 2020 – July 2024

B.Tech Computer Science and Engineering

GPA: 9.23

SKILLS

Languages: Python - Data Structures and Algorithms, React, C++, C#, SQL, JavaScript, Postgres, PyTorch

Technologies: Google Cloud Platform, Microsoft Azure, Docker, OpenAI Playground, Hugging Face

Frameworks: Machine Learning, NLP LLMs, Deep Learning, Generative AI, Diffusion Models

EXPERIENCE

Unremot, Chennai

Feb 2024 – Oct 2025

AI Engineer

- Developed AI-bots and Co-pilots for The Wadhvani Foundation's Ed-Tech platforms and collaborated in creating, training, and testing two AI Ed-Tech platforms tailored for student entrepreneurs and SME business owners.
- Designed advanced prompting strategies, including COT, TOT, ReAct, Reflexion, Self-consistency and multi-agent methods, improving system robustness and preventing prompt hacking almost 95%.

SOUL AI, Chennai

Sept 2023 – Jan 2024

AI Engineer

- Trained an AI tutor model using Reinforcement Learning to generate personalized, context-aware responses.
- Achieved over 96% accuracy in generating human-like responses through reward modeling and fine-tuning.

PROJECTS

Human vs AI Code Detection | *PyTorch, GraphCodeBERT, Contrastive Learning, Python*



- Trained a BERT-style classifier using contrastive learning on a custom dataset of 600k commits from GitHub unified diffs to predict AI vs human code, tested across 1,700 repositories spanning both before and after the LLM era.
- Evaluated on out-of-distribution data, achieving 93% accuracy and an F1 score of 0.979 with a 400M-parameter model.

SLIM-DIFF: Diffusion-Based Code Editor | *PyTorch, Diffusion LLMs, QLoRA, LoRA, Hugging Face*



- Built a surgical line-level code repair pipeline using diffusion language models for localized Python bug fixing, benchmarking DiffuCoder-7B, Stable-DiffCoder-8B, Qwen2.5-Coder, and Seed-Coder under a unified evaluation framework.
- Fine-tuned diffusion-based code infill models with custom masking and denoising objectives, improving repair accuracy to nearly 96% while increasing runnable and syntax-valid program generations compared to base diffusion checkpoints.

Arivian - Research Assistant | *React, Tailwind, Vector DB, Postgres, Python*



- Built a collaborative research papers platform using FASTAPIs and Llama 3.1 with Supabase integration.
- Implemented FAISS in a RAG pipeline, using the mpnet-base V2 sentence transformer to generate high-quality embeddings for efficient and accurate semantic retrieval and chat with research papers using OpenSource API.

3D Generation using Vision Transformers | *Python, PyTorch, DINO-V2, Diffusers*



- Built a Vision Transformer pipeline for developing high-fidelity 3D texture generation from multi-view 2D images.
- Used NeRF, GANs and VAE with improved CNN for volumetric reconstruction and trained on Objaverse dataset.

Dynamic AI Gaming with Large Language Models | *LLMs, RAG, Vector Databases, Pytorch*



- Created a real-time NPC dialogue generation using LLMs with low-latency inference pipeline using Opensource LLMs.
- Integrated external text retrieval and high-dimensional vector search to support context-aware quest generation and gameplay logic conditioned on evolving player and world state.

Optimized Image Generation Pipeline | *PyTorch, Diffusion Models, SDXL, Whisper V2 Large*



- Deployed an enhanced image generation pipeline for low latency and reduced memory footprint.
- Used float16 inference, FlashAttention, and accelerated samplers to achieve 60% faster and detailed results.

CERTIFICATES & PUBLICATIONS

Reinforcement Learning, MCP Agents, 3D Diffusion models

Hugging Face

Type Sculpt: Text-to-3D Generation using Adaptive Attention Mechanism

IRCCTSD'24